

Pavan Kalyan Varikuppala

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EDUCATION

Osmania University, Hyderabad

B.Sc in Data Science

2023 – 2026

GPA: 8.4/10

EXPERIENCE

Full Stack Developer Intern

Mar 2026 – Present

Gaara AI (Nirmaan Pre-Incubator, IIT Madras)

- Architected production-grade full-stack systems on **FastAPI**, **React.js**, and **PostgreSQL** (SQLAlchemy ORM), enforcing modular service boundaries and async I/O for concurrent workloads.
- Containerized services with **Docker** and orchestrated zero-downtime deployments on **AWS EC2**, leveraging **S3** for object storage and static asset distribution.
- Hardened network infrastructure with **Nginx** reverse-proxy, **Cloudflare** DNS/WAF, DuckDNS, and **Certbot** for automated TLS certificate provisioning and renewal.
- Profiled and optimized backend hot paths — query indexing, connection pooling, and response caching — to sustain low-latency throughput under real-world traffic.

PROJECTS

Webhook Mock Studio | *Dynamic API Platform* | *Django, DRF, React, JWT*



- Designed and shipped a production-deployed API mocking platform supporting on-demand endpoint generation with user-defined URLs, HTTP methods, and JSON payloads — live at webhook-mock-studio.vercel.app.
- Engineered a real-time observability dashboard surfacing request logs, latency telemetry, and traffic patterns to accelerate developer debugging cycles.
- Implemented stateless **JWT** authentication with silent access-token rotation and automated refresh cycles, eliminating mid-session re-authentication friction.

Secure Task Manager | *Full Stack Application* | *Django, DRF, React*



- Engineered **REST APIs** enforcing strict per-user data isolation through row-level access control and authorization middleware.
- Implemented protected SPA routing with a *JWT access/refresh-token* architecture and silent token rotation, mitigating session-hijack attack surface.
- Optimized ORM access patterns and indexing strategy to handle high-frequency CRUD operations with minimal query latency.

LLM-Powered AI Chatbot | *CRTOperations* | *Python, Streamlit, LangChain*



- Developed a context-aware conversational AI agent capable of resolving multi-turn, domain-specific queries with persistent dialogue state.
- Integrated **LangChain** orchestration with LLM APIs to manage long-context conversation memory, prompt chaining, and intent-based intelligent routing.
- Delivered a responsive Streamlit interface with conversational state management and instant message rendering for an end-user-ready experience.

TECHNICAL SKILLS

Languages: Python, C

Backend & APIs: Django, Django REST Framework, FastAPI, REST APIs, WebSockets

Frontend: React.js, JavaScript, HTML5, CSS3

Databases: PostgreSQL, MySQL, SQLite

Cloud & DevOps: AWS (EC2, S3), Docker, Nginx, Cloudflare, DuckDNS, Linux, Git

AI / Data: LangChain, LLM APIs, NumPy, Pandas

CERTIFICATIONS & ACHIEVEMENTS

Certifications: C Programming, MS Office